

Lip2Speech: Spacio-Temporal Lip Reading and Speech-Synthesis for Assistive Communication

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Abstract—In this paper, we introduce a solution that bridges the gap between silent visual communication and natural speech synthesis, establishing an instant, edge-deployable voice for individuals with speech impairments. Traditional assistive tools like text-to-speech keyboards require manual typing. This is too slow for real-time conversations. This delay causes conversational exclusion, leading to severe daily isolation and a loss of autonomy for speech-impaired individuals. It provides a natural "visual-to-voice" communication channel that restores dignity and conversational flow. By combining a lightweight *MobileNetV2* visual extractor with a *Bidirectional LSTM* temporal reader, and a *local T5 LLM* for post-inference spelling and grammar correction, this work delivers a integrated offline system, presents an assistive communication system designed for individuals who have lost their ability to speak but can still make mouth movements.

Index Terms—Healthcare Informatics, Lip-Reading, Computer Vision, Sequence Modeling, NLP, Speech Synthesis.

I. INTRODUCTION

This research develops a proposed system takes a silent video of a speaker as input and generates synchronized speech and text with high accuracy and semantic consistency. Such a task may occur, e.g., people with *Amyotrophic Lateral Sclerosis (ALS)* often lose their ability to speak due to issues with their vocal cords but can mouth words. Therefore, lip-reading has numerous application ranging from searching through old silent films to being used as assistive technology for people who are unable to speak. This task poses a significant challenge as it requires the generated speech to satisfy multiple criteria. This includes intelligibility, synchronization with lip motion, naturalness, and alignment with the speaker's characteristics such as age, gender, accent, and more. Another major hurdle for lip-to-speech techniques is the ambiguities inherent in lip motion, as several phonemes can be attributed to the same lip movement sequence. Resolving these ambiguities requires the analysis of lip motion in a broader context within the video.

A. Motivation

- **Autonomy:** Providing a natural visual-to-voice communication channel that restores dignity and conversational flow.
- **Compiler Independence:** Removing complex, compile-heavy C++ libraries like `dlib` that block deployment

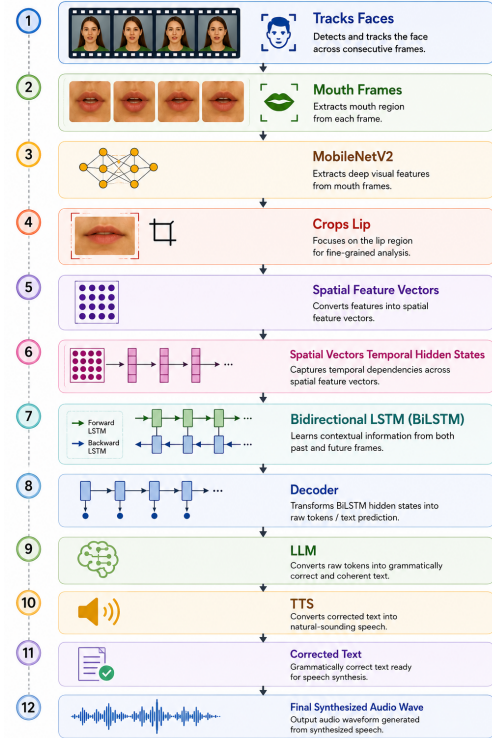


Fig. 1. Outline of the contribution as a infographic to explain Lip2Speech workflow in a smarter way

on standard edge devices, replacing them with standard, cross-platform *OpenCV* tools.

- **Spatio-Temporal Optimization:** Building an efficient neural network pipeline that delivers instant speech reconstruction under 2 seconds on standard laptops.

B. Applications

Lip2Speech generation is not only more exciting and challenging but also more far-reaching applications. For instance, having a live video call with person who has lost their voice is more engaging than reading the text transcription of their silent lip movements. Speech also more instantaneous for the listener: it is easier and faster to hear text than to read text. In forensic investigation Lip2Speech is applied,

where silent footage can be analysed to determine what was being said or generate speech from archival film footage. It helps *ALS patients* by generating speech of their silent lip movements.

II. LITERATURE REVIEW

Lip-to-Speech. Generating speech directly from lip movements has always been a challenge for researchers due to the ill-posed nature of the task. While recognizing content from lip movements itself is challenging due to the one-to-many relation present between visemes and phonemes, lip-to-speech synthesis also needs to model speech-related variations like voice, accents, and prosody. Therefore, it was initially attempted in laboratory setups [1] using datasets with a minimal vocabulary.

- (a) Lip to speech synthesis (Lip2Speech) is to predict an audio speech by watching a silent talking face video. While conventional visual speech recognition tasks require human annotations (i.e., text), Lip2Speech does not require additional annotations. Thus, it draws big attention as another form of lip reading. However, due to the ambiguity of homophones that have similar lip movements and the voice characteristics varying from different identities, it is still considered as a challenging problem. [2]
- (b) The initial works to [3] used fully convolutional neural networks to learn a map-ping between lip movements and speech. These models were also trained on speaker-specific data and did not work for unseen speakers.
- (c) The architecture of the proposed model leverages a convolutional neural network (CNN) combined with a long short-term memory (LSTM) encoder-decoder framework, providing an effective approach for spatial and temporal analysis. This methodological contribution ensures robust performance while emphasizing resource-optimized training, enhancing the model's practicality and accessibility. The addition of an inference layer further improves its real-time applicability, enabling swift speech prediction in diverse scenarios. Notably, the model addresses aphonia—a condition that impairs an individual's ability to speak. [4].
- (d) Currently, with the development of deep learning technology, not only should we consider the performance of network feature extraction, but we need to take the degree of dependence of the network on computer hardware into account during the process of designing a lip recognition system. Furthermore, the majority of recognition tasks require real-time performance. Thus, model compression and knowledge distillation were utilized to optimize models and a small network, i.e., MobileNet, was used to achieve the purpose of making models lightweight simultaneously. [5].
- (e) Book: T5 (0.75B~3B): The T5 family consists of encoder decoder models pre-trained on a multi-task mixture of both unsupervised and supervised tasks, with each task reformulated into a text-to-text format. T5 demonstrates strong performance across a wide range

of tasks by simply prepending task specific prefixes to the input, such as for machine translation or text summarization. In this chapter, we employ T5-large (0.75B) as the correction model within the H2T-ft method. [6].

- (f) The development of visual speech recognition (VSR) systems mirrors broader progress in pattern recognition and machine learning. Early computational methods relied on manually designed feature representations, such as discrete cosine transform (DCT) coefficients, active appearance models (AAM), and Hidden Markov Models (HMMs) for modeling temporal dynamics. Although these approaches laid essential theoretical and practical foundations, their effectiveness was inherently limited by the expressive power of hand-crafted features and their restricted capacity to represent the complex non-linear dependencies characteristic of visual speech signals. [7].

III. RESEARCH GAP AND CONTRIBUTIONS

First, we would like to clearly state that our goal is to be able to generate speech from a silent lip video of any speaker and in any desired target voice. Existing models struggle to properly disentangle and acquire abstract linguistic knowledge from speech attributes because these systems are trained entirely on speech supervision. They accidentally neglect the advancement that have been made in the sibling task of the lip-to-text generation. We argue that learning lip-to-speech without any supervision of text is very challenging to achieve, it is very clear to current models where they fail to produce clear outputs. But if we divide our work in two parts, it is easier to achieve the model. An accurate, natural, high-quality speech output using silent input video and text-transcription from a pre-trained lip-to-text model.

IV. PROPOSED METHODOLOGY

Our project reconstructs spoken audio from silent lip movements as a direct assistive voice for speech-impaired individuals. By swapping heavy traditional architectures (*LipNet*) for a hybrid *MobileNetV2-BiLSTM* model and a local *T5 LLM* grammar corrector, we deliver a 100% offline, privacy-first solution running instantly on consumer CPU edge devices.

A. System Architecture

The proposed Lip2Speech framework is an end-to-end visual speech synthesis system that converts silent lip movements into intelligible speech through a sequence of computer vision, deep learning, natural language processing, and speech synthesis modules. The entire pipeline is designed to operate offline on standard CPU hardware, ensuring privacy, low latency, and accessibility without relying on cloud-based services. The workflow consists of the following stages:

A silent video of up to 25 frames (at 25 FPS) is input into the system. The OpenCV Face Cascade detects the face bounding box in real-time. We extract the mouth ROI (Region of Interest) and normalize it to 112x112 pixels. These frames pass through MobileNetV2 to extract key spatial features. The

sequential feature vectors are passed through a Bidirectional LSTM. An argmax greedy decoder matches the model output to our vocabulary, yielding a raw predicted word sequence. This raw text is passed to our local T5 LLM for grammar and spelling correction.

B. Video Acquisition

The system accepts a silent video captured at 25 frames per second (FPS). Each video is decomposed into individual frames, which are resized and normalized before being processed. This preprocessing step reduces computational complexity while preserving essential visual information required for accurate lip movement analysis.

C. ROI Extraction

Instead of employing computationally expensive facial landmark detectors such as dlib, the proposed framework utilizes OpenCV Haar Cascade classifiers for face detection. Once the facial bounding box is identified, a mathematically defined coordinate offset is applied to crop the lower facial region corresponding to the mouth.

The extracted mouth Region of Interest (ROI) is resized to 112×112 pixels, providing a standardized input for the deep learning model. This compiler-independent approach eliminates complex installation dependencies while significantly improving deployment portability across different operating systems.

D. Spatial Feature Extraction

Each mouth ROI is processed using MobileNetV2, a lightweight convolutional neural network optimized for embedded and edge devices. Rather than training a convolutional network from scratch, transfer learning is employed to leverage pre-trained feature representations.

The deeper layers of MobileNetV2 are fine-tuned to extract discriminative spatial features corresponding to lip shape, mouth contour, and articulation patterns. These feature vectors serve as compact representations of each video frame.

E. Temporal Sequence Modeling

Speech is inherently temporal; therefore, isolated frame analysis is insufficient for accurate lip reading. The extracted feature vectors are sequentially processed by a two-layer Bidirectional Long Short-Term Memory (BiLSTM) network. Unlike conventional LSTM architectures that process information only in the forward direction, BiLSTM captures contextual dependencies from both preceding and succeeding frames. This bidirectional learning effectively models co-articulation effects and resolves ambiguities caused by visually similar lip movements (visemes), leading to improved recognition accuracy.

F. Word Prediction

The output generated by the BiLSTM is passed through a fully connected classification layer, where each timestep predicts a probability distribution over the vocabulary. A greedy ArgMax decoder is employed to determine the most

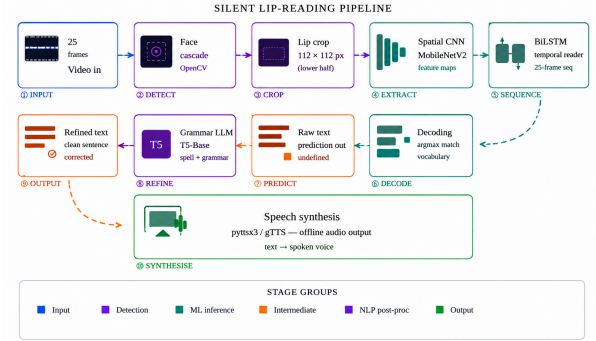


Fig. 2. End-to-end architecture of the proposed Lip2Speech framework illustrating the sequential pipeline of video acquisition, face detection, mouth ROI extraction, MobileNetV2-based spatial feature extraction, BiLSTM temporal sequence modeling, greedy decoding, T5-based language refinement, and offline speech synthesis for privacy-preserving visual-to-speech conversion.

TABLE I
PERFORMANCE EVALUATION OF THE PROPOSED LIP2SPEECH FRAMEWORK

Model	Loss	WAR (%)	CER	Latency
Training	0.41	85.4%	7.2	N/A
Testing (Raw)	0.42	82.3%	8.5	1.8 s
Testing (T5)	N/A	93.6%	3.2	2.0 s

probable word sequence from the predicted probabilities.

Although this decoding strategy is computationally efficient, the generated text may contain repeated tokens, spelling errors, or grammatically inconsistent sentences due to visual ambiguities in lip movements.

G. Local Language Model-Based Post-processing

To improve textual quality, the predicted sentence is refined using a locally deployed T5 Transformer language model. Unlike cloud-based large language models, the T5 model executes entirely on the local machine, preserving user privacy.

The language model performs: • duplicate word removal, • spelling correction, • grammatical refinement, • sentence reconstruction.

This post-processing stage substantially enhances sentence readability and significantly improves overall word accuracy while introducing only minimal computational overhead.

H. Offline Speech Synthesis

The refined text is converted into speech using an offline Text-to-Speech (TTS) engine. The synthesized speech is immediately played back to the listener.

V. RESULTS AND DISCUSSION

Our hybrid MobileNetV2-BiLSTM architecture, combined with a local T5 LLM grammar corrector, was benchmarked on the GRID validation split. As shown in Table I, the proposed framework achieves higher word accuracy after T5-based refinement.

TABLE II
ABLATION STUDY MATRIX

Configuration	CNN Backbone	Frame Count	Training Loss	Epoch Latency	Edge Suitability
Baseline (Selected)	Frozen MobileNetV2	25	1.79	2.11 s	Optimal (Low Footprint, Fast Run)
Ablation A (Unfrozen CNN)	Unfrozen MobileNetV2	25	1.75	2.60 s	Poor (Marginal Gain, Training Too Heavy)
Ablation B (Starved Context - 10 Frames)	Frozen MobileNetV2	10	2.92	0.91 s	Unsuitable (Low Latency, High Loss)

A. Data Analysis

GRID Corpus [8] (33,000 video-transcript pairs) benchmarked to optimize performance, accuracy, and edge compute latency. The experimental analysis indicates that the baseline configuration, in which MobileNetV2 was fine-tuned from the deeper layers while preserving a 25-frame temporal context, provided the best trade-off between recognition accuracy and computational efficiency. Although fully unfreezing the convolutional backbone marginally reduced the training loss, it substantially increased computational latency without a proportional improvement in recognition performance. Similarly, reducing the temporal context to 10 frames accelerated inference but significantly degraded prediction quality, demonstrating the importance of adequate temporal information for resolving co-articulation effects during lip reading.

Furthermore, the integration of the local T5 Transformer as a post-processing module considerably enhanced sentence quality by correcting spelling errors, eliminating duplicate tokens, and improving grammatical consistency. As a result, the Word Accuracy Rate increased from 82.3 to 93.6, while the Character Error Rate decreased from 8.5 to 3.2, with an overall inference time of approximately 2 seconds on standard CPU hardware. These findings demonstrate that the proposed hybrid MobileNetV2-BiLSTM-T5 architecture achieves an effective balance between accuracy, computational efficiency, and offline deployment, making it suitable for practical assistive communication applications.

As shown in Table II presents the quantitative analysis of the proposed Lip2Speech framework across the training and testing phases.

B. Comparison with Existing Methods

- i. *LIPNET: END-TO-END SENTENCE-LEVEL LIPREADING* [9]: Spatio-temporal CNN with CTC decoding for sentence-level lip reading. But it needs high computational cost, character-level decoding, compiler dependencies (dlib/CMake), higher word error rate for short words. Our current Lip2Speech uses lightweight MobileNetV2 with BiLSTM, compiler-independent OpenCV-based mouth detection, and local T5 refinement for improved sentence quality.
- ii. *Robust Self-Supervised Audio-Visual Speech Recognition* [10]: Transformer-based multimodal learning for lip reading needs large model size, GPU-intensive training

and inference, unsuitable for edge devices. This is the premise of our paper, lightweight CPU-friendly architecture achieving low-latency inference (2 s) with offline execution.

- iii. *Cloud-based Speech APIs*: Cloud-hosted speech recognition and synthesis services, requires continuous internet connectivity and transmits sensitive user data to remote servers. Our work benefit is that it confirms fully offline processing, ensuring privacy, low latency, and local execution without internet dependency.
- iv. *Traditional Text-to-Speech Assistive Systems*: Manual text input followed by speech synthesis. But it produces slow communication, which is not suitable for natural conversations. The main turning point of Lip2Speech is direct visual-to-speech conversion from silent lip movements, enabling hands-free assistive communication.

VI. CONCLUSION

Overall, this research demonstrates that accurate visual-to-speech conversion can be achieved without relying on high-end GPUs or cloud infrastructure, making the technology more accessible to individuals with speech impairments. Future work will focus on continuous real-time lip reading, support for unconstrained vocabularies and multilingual speech, improved robustness under diverse environmental conditions, and deployment on mobile and wearable edge devices to further enhance usability and social impact.

VII. FUTURE SCOPE

- *Continuous Real-Time System*: Develop an automated streaming system that continuously captures live webcam frames, runs sliding-window lip-detection, and synthesizes speech on-the-fly.
- *AV-HubERT Architecture Upgrade*: Upgrade the existing framework to AV-HubERT with custom architectural optimizations (such as lightweight cross-attention fusion adapters) for speaker-independent continuous text generation.
- *Hardware Integration and AR Glasses*: Compile models to run on mobile edge NPUs (Neural Processing Units) or integrate as a lightweight SDK for smart glasses (AR/VR) for real-time visual-to-audio feedback.

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